

**Rochester Institute of Technology
Department of Computer Engineering
CMPE-630 Digital IC Design
Laboratory Exercise 1
Introduction to Cadence Design Systems
Due: In-Lab**

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BEFORE YOU GET STARTED

Cadence, and perhaps many other tools that you will encounter, are all very *complex*. These tools are industry-standard, and they offer tons of features at the cost of being relatively **complex** compared to the other software you may have worked with in the past (e.g. Vivado).



These tools are difficult to work with and they require a lot of attention and patience. You will encounter bugs; sometimes these tools may present warnings that are confusing and not explicit, or may not be real. Pay close attention, use common sense and critical thinking skills to overcome the difficulties encountered using these tools. i.e. **You must read through everything carefully**, or these tools will surely lead you down the wrong path!

Keep the following in mind as you work through these labs.

- **NEVER** COPY CADENCE LIBRARY FILES FROM THE COMMAND LINE OR FILE MANAGER
- IF ANY OF YOUR CADENCE FILES IS LOCKED, CLOSE THE TOOL AND EXECUTE THE FOLLOWING COMMAND:

```
/classes/ce620/bin/unlock_cadence.csh
```

- **Hotkeys:** You should learn how to use the hotkey for better productivity. You can access a list of hotkeys (also known as bindkey) from CIW (the first window that pops up after you launch virtuoso). In CIW, select the menu option:

```
Options → Bindkey
```

Select a category (e.g. layout) and a list of current hotkeys (bindkeys) will be shown for it

Have fun and good luck!

1 Introduction

Welcome to CMPE 530/630, Digital IC Design! The laboratory portion of this class will introduce you to the version 6.x IC Custom, full-custom Integrated Circuit design environment using Cadence Design Systems. You will target a generic 45 nm process design kit (PDK) and work through the various steps in circuit design flow to create a custom standard cell library consisting of several CMOS logic gates. These tools offer schematic capture, pre-layout circuit simulation, full custom layout, design rule checks (DRC), layout vs. schematic (LVS), parasitic extraction, and post-layout circuit simulation with extracted parasitics, which you will learn how to use throughout the course.

Disclaimer: The instructions for this lab were made with an older version of Cadence Virtuoso.

For this lab, we will focus on:

- Getting started with the setup of the Cadence environment and getting familiar with its tools
- Creating a schematic of a CMOS logic gate

There is **NO lab report for this lab.**

The deliverable is the worksheet with all of the signoffs and questions answered

2 Starting a Session

If you are working in ICE Lab (GLE-3159) or VLSI Lab (GLE-3149), just log onto one of the Linux workstations using your RIT login and skip to section 3 - SETTING UP THE CADENCE ENVIRONMENT

Remote Access

For working at home or outside of the lab, remote access to the Linux Workstations (in GLE-3159 or GLE-3149) is **preferably performed using a FastX client**. Other applications such as MobaX-term or PuTTY with Xming also work, but they are usually slower than FastX.

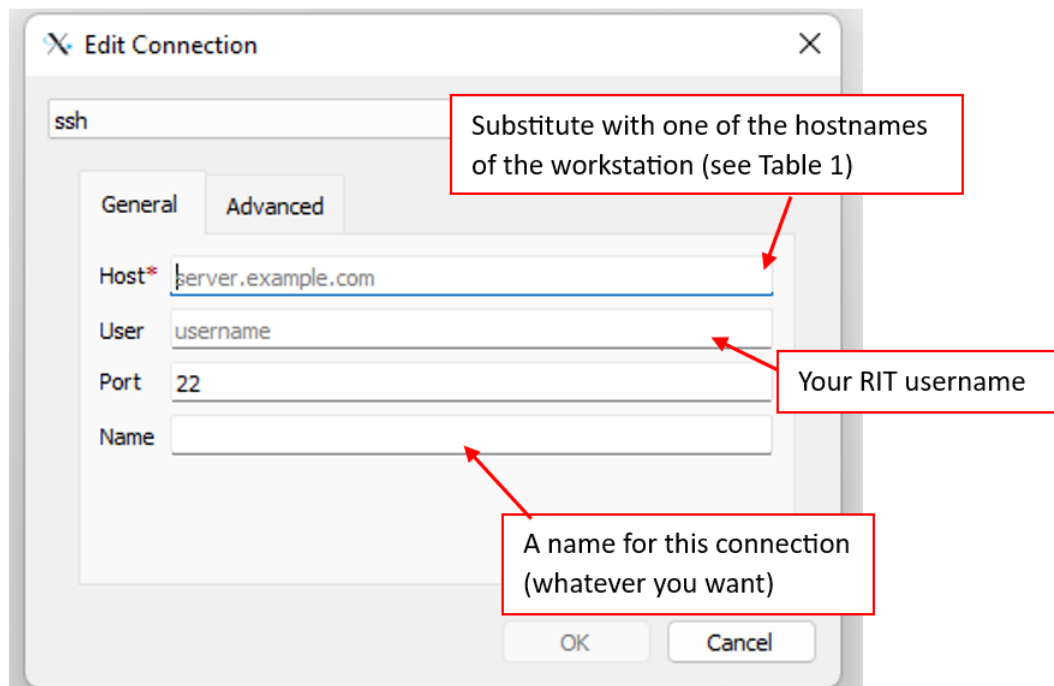
There are two ways to use FastX to access a Linux Workstation:

1. Web browser

- a. Review the web browser instructions on this web page
 - * **Note:** Please use **3300** as the port number instead of **3443**.
- b. Substitute **computername** with one of the hostnames of the workstation (see Table 1)

2. FastX Client (**THIS IS THE PREFERRED METHOD**)

- a. Install FastX client from here
- b. Once installed, create a new connection by hitting **File → New**



- c. Double-click on the newly created connection to start SSH

- d. Afterward, hit the plus button to start a new session. Choose GNOME, and then hit the play button to start the session

Table 1: Linux Workstations

ICE Lab GLE-3159-WS01 (up to WS32)	Example Hostname: GLE-3159-WS09.nx.rit.edu
VLSI Lab GLE-3149-WS01 (up to WS16)	Example Hostname: GLE-3149-WS15.nx.rit.edu

Make sure to log out and terminate the ssh session when you are done.

3 Setting up the Cadence Environment

◇ You only have to do this once ◇

Download `cdssetup` from MyCourses and place it in your home directory. Then, type:

```
chmod +x cdssetup
./cdssetup
```

You should now have a new directory called `cmpe630` in your home directory which will eventually be populated with files and directories related to your lab work.

Now, log out of your current session, then log back on.

Once you are logged back on, change to your project work directory by executing the following command and hitting enter after.

```
cd cmpe630
```

Now start the Cadence tools by executing the command:

```
virtuoso &
```

A new window called CIW¹ pops up, shown in Fig.1.

Keep the CIW window visible at all times as it maintains a log of what the Cadence software is doing at any point in time, including **warnings** and **errors**.

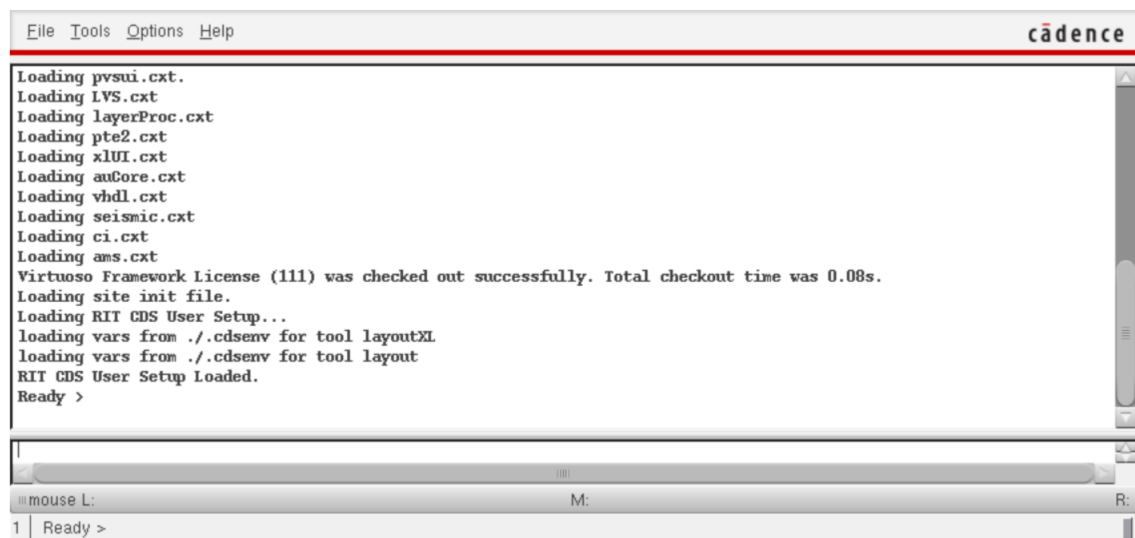


Figure 1: Cadence Virtuoso CIW

Once the CIW starts, the Library Manager should also open as shown in Fig.2.

¹CIW stands for Command Interpreter Window

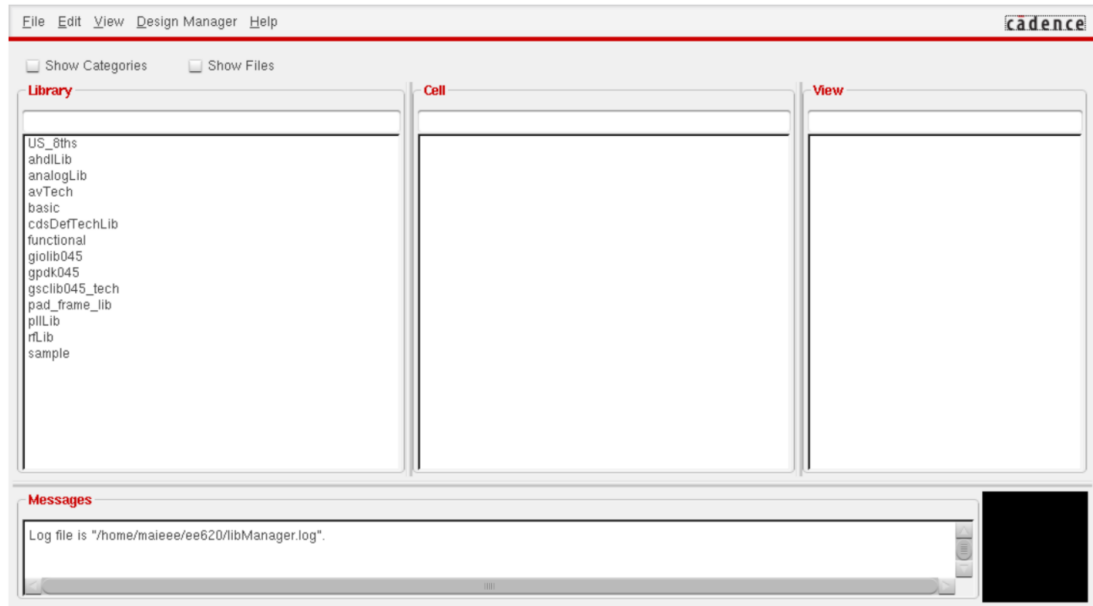


Figure 2: Cadence Virtuoso CIW

4 Creating Your Own Cell Library

◇ You only have to do this once ◇

Note: Although the Library Manager should be open, if you **happen to close it** you can open it from the CIW window menu. Select **Tools → Library Manager** and the window shown in Fig.3 will show up.

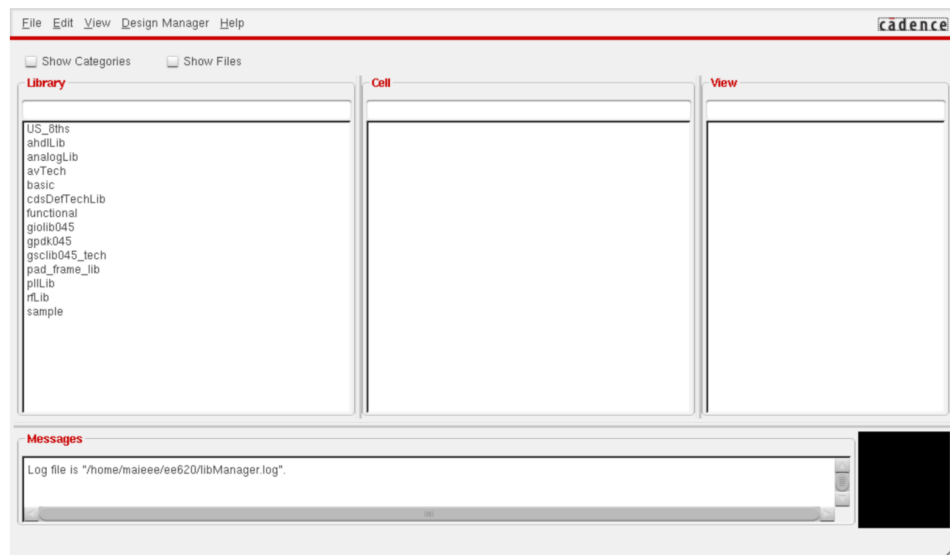


Figure 3: Cadence Virtuoso Library Manager

To create your own new library, select **File → New → Library**.

A new window will pop up to insert a name for the library (see Fig. 4). Name it using your username initials “fml” (first, middle, last) i.e. **fml.lib**. For consistency, please use all lowercase and follow these instructions so the instructor and the TA’s can help you with future labs!

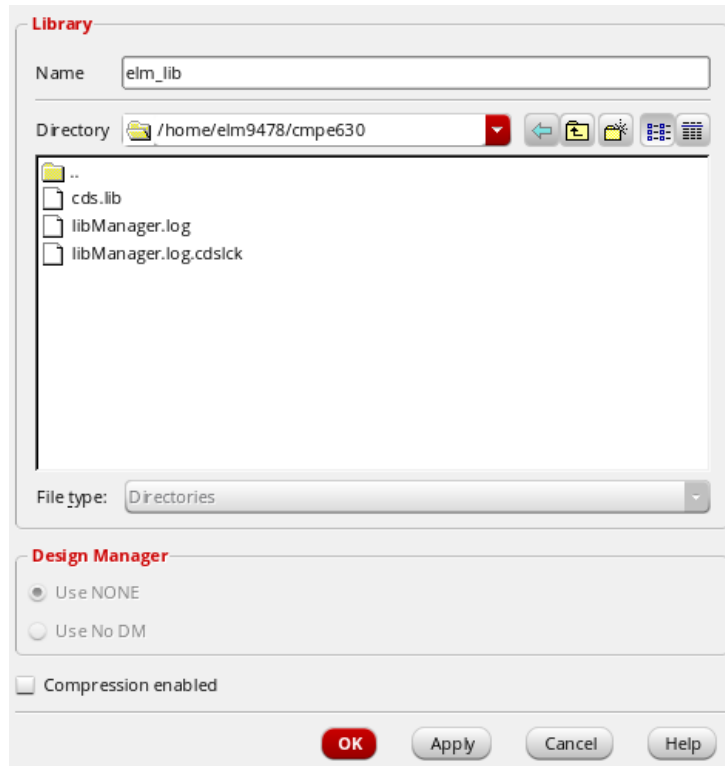


Figure 4: Create New Library Form

Make sure the directory shows up as **/home/abc1234/cmpe630**. Then **Click OK**. Then, A new window will pop up. Select **Attach to an existing technology file** (see Fig. 5).

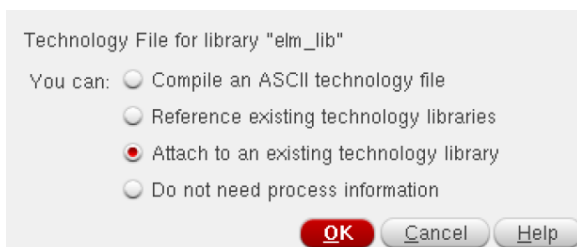


Figure 5: Attach To an Existing Technology File



Figure 6: Technology Library

Click **OK**. A new window pops up shown in Fig. 6. Then select the technology file for the Generic 45 nm standard cell technology library, click **gsclib045_tech**.

Click **OK**. You'll be returned to the Library Manager Window. If the attachment is successful, you should see these messages logged in the CIW window (see Fig. 7).

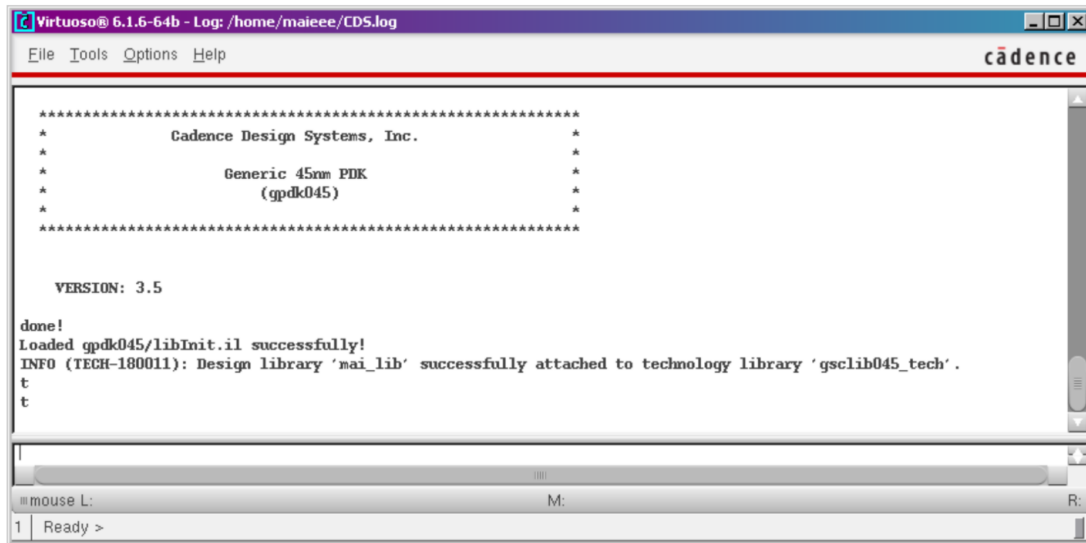


Figure 7: Technology Library Attachment CIW log

In the left-hand column of your library Manager, you'll see your new library **fml_lib** (see Fig. 8).

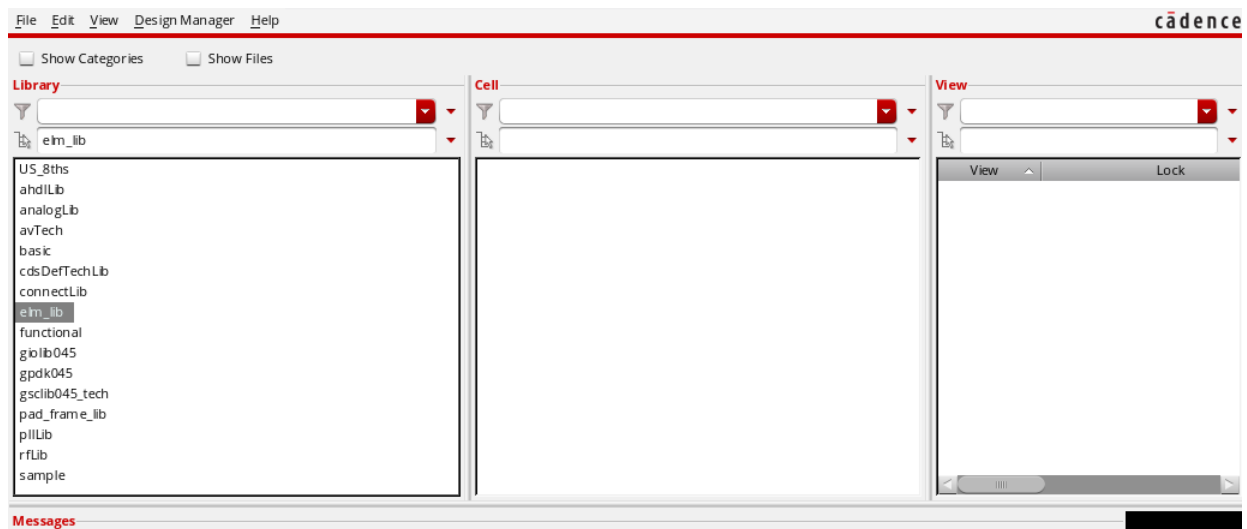


Figure 8: Library Manager Showing New Library

This completes the creation of your new library. You can check the properties of your new library by right clicking on your library and select Properties; the following window appears. Your library properties should be similar (see Fig. 9).

The screenshot shows a window titled "Library Property Editor@gle-3159-pc25". It contains three main sections: "Library", "UNIX Permission Mode", and "Properties".

- Library**: Fields include Name (elm_lib), Owner (elm9478), Group (5003), Last Modify (20 20:23:35 2021), Read Path (/home/elm9478/CMP), and Write Path (/home/elm9478/CMP).
- UNIX Permission Mode**: Fields include Owner (rwx), Group (rwx), and Other (---).
- Properties**: Fields include cdfData (type "schematic" instDisplayMode "instName")) and techLibName (gscLib045_tech).

At the bottom, there are buttons: OK, Cancel, Apply, Add, Delete, Modify, and Help.

Figure 9: Library properties Form

All your cells and work will be created in your new library. You do not need to create a new library every time you start the Cadence tools!

Now close the Library Manager; close the Cadence tools and then log off.

Then log on and return to your cmpe630 directory and **restart** the Cadence tools.

REMINDERS:

- Always start the Cadence tools using the **virtuoso &** command from your cmpe630 directory and NOT from your home directory.
- Always properly close the Cadence tools before you log off, otherwise you WILL lose your work and/or potentially lock your cells and not be able to access them!

5 Creating A New Cell

Reminder: *'fml' is an example for first, middle, last name. Please use your initials!*

In the Library Manager window, select your library **fml_lib** and then select **File → New → Cellview**

As shown in Fig. 10 below select **fml_lib** and give your first schematic the name of **FML_SCHX1**

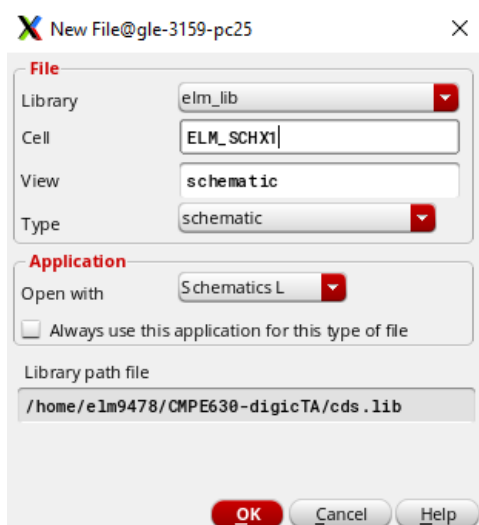


Figure 10: Create a New Cellview Form

Click OK.

A warning window may appear indicating that the license for Schematics L is not available, and do not wish to use Schematics CL instead. Click **Always** and a new window pops up (see Fig. 11). This is the Composer schematic capture window. On the top are text based menus, and icons for the most often used commands.

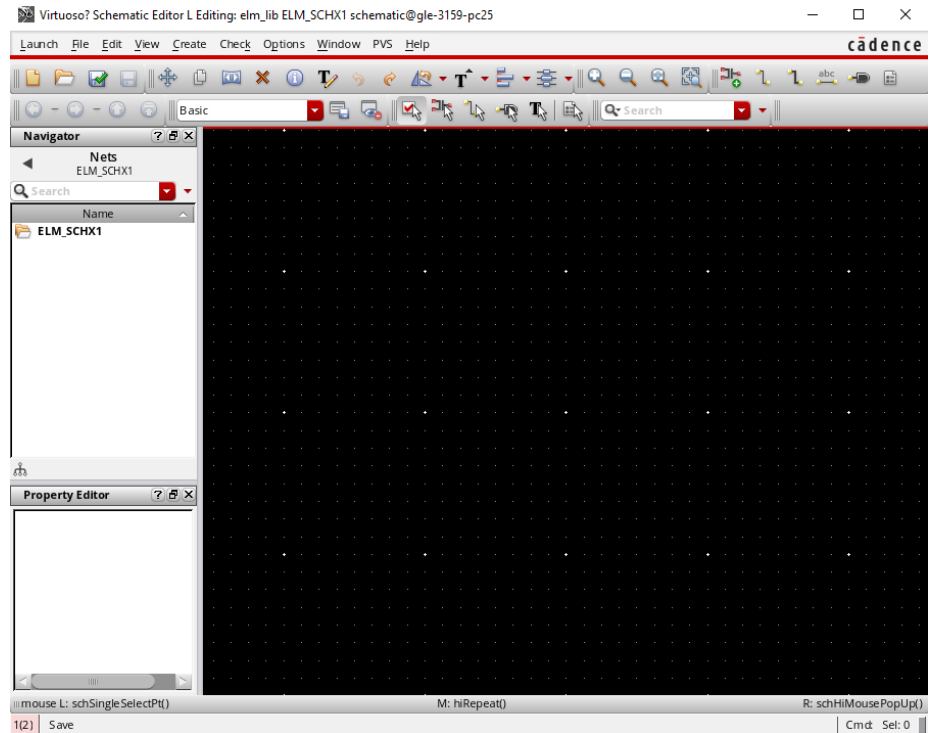


Figure 11: New Composer Schematic Window

Select **Create Instance** from the icons or hit **Create → Instance**. A new window (see Fig. 12) should then pop up.

Figure 12: Add Instance Form

Click **Browse** to select the library, another new window should then pop up (see Fig. 13). Select **gpdK045** as the library, cell **nmos1v**, and **symbol** view.

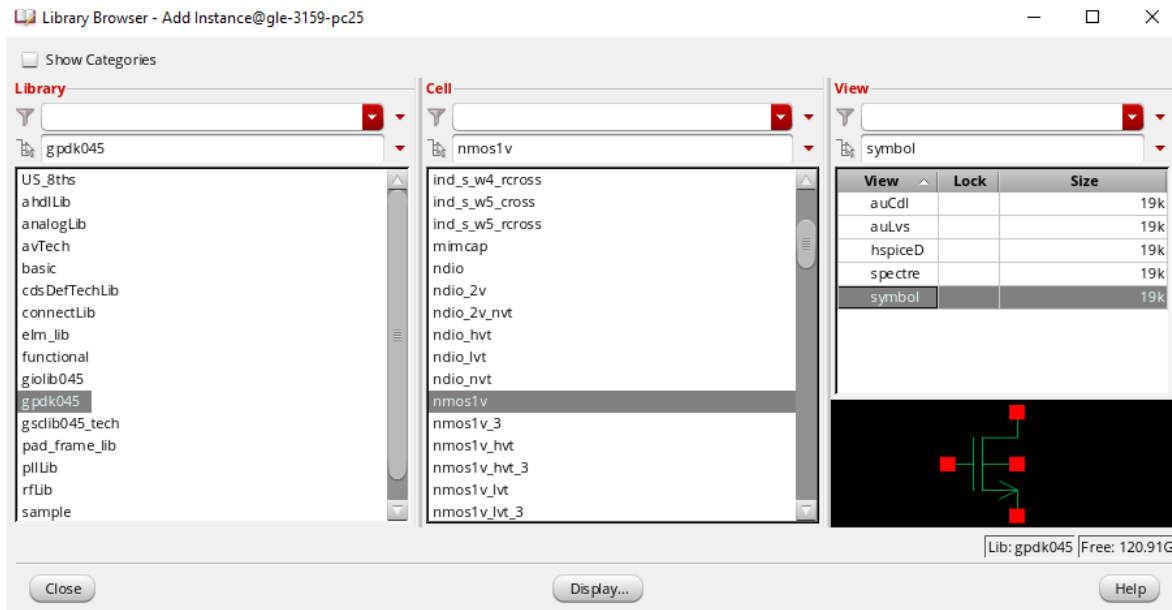


Figure 13: Library Manager

Once you have selected the symbol view, the following window pops up (see Fig. 14). Adjust the transistor parameters Length (the channel length) and Total Width **ONLY** as shown in Fig. 14. **DO NOT ADJUST ANY OTHER PARAMETERS!**

The **Total Width** should automatically update once you hit the tab key.

The "Add NMOS Transistor Instance Form" window shows the configuration for a new transistor instance.

- Library:** gpd045
- Cell:** nmos1v
- View:** symbol1
- Names:** (empty field)
- Add Wire Stubs at:**
 - ☒ all terminals
 - ☐ registered terminals only
- Array:** Rows: 1, Columns: 1
- Buttons:** Rotate, Sideways, Upside Down
- Model Name:** g45n1svt
- Multiplier:** 1
- Length:** 45n M
- Finger Width:** 240n M
- Total Width:** 240n M
- Fingers:** 1
- Folding Threshold:** 10u M
- Diff Cont:** ☒
- Gate Connection:** None
- Use DRM Rules:** Minimum
- S/D Metal Width:** 60n M
- Switch S/D:** ☐
- Bodytie Type:** None
- Buttons:** Hide, Cancel, Defaults, Help

Figure 14: Add NMOS Transistor Instance Form

Click **Hide** and in the main capture window you'll see the yellow symbol of a 4-terminal NMOS tied to the cursor. Place it anywhere by clicking on a suitable location in the schematic canvas.

Once finished placing your transistor, press ESC. A NMOS should then be placed on the main capture window.

You do:

- Add 2 more of these (see Fig. 15), you should have a total of **3** NMOS with this same sizing

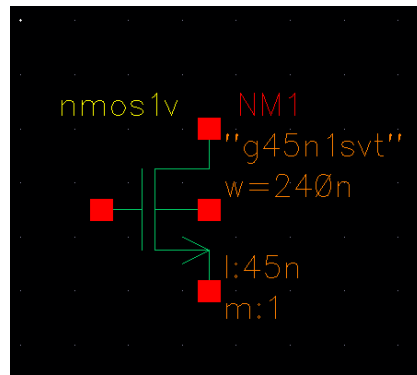


Figure 15: Add NMOS Transistor Instance Form

- Do the same for a **pmos1v** transistor with **length (45nm)** and **total width (480nm) for two** and **one with total width of (240nm)**. You should end up with an arrangement like the one below (see Fig. 16). *You will learn about sizing in later lectures but for now we are giving you the correct sizing.*
- **Note:** *You can quickly change the properties of the instance by either right clicking the selected instance or pressing q*

From the menu or icons select **View → Zoom To Fit**

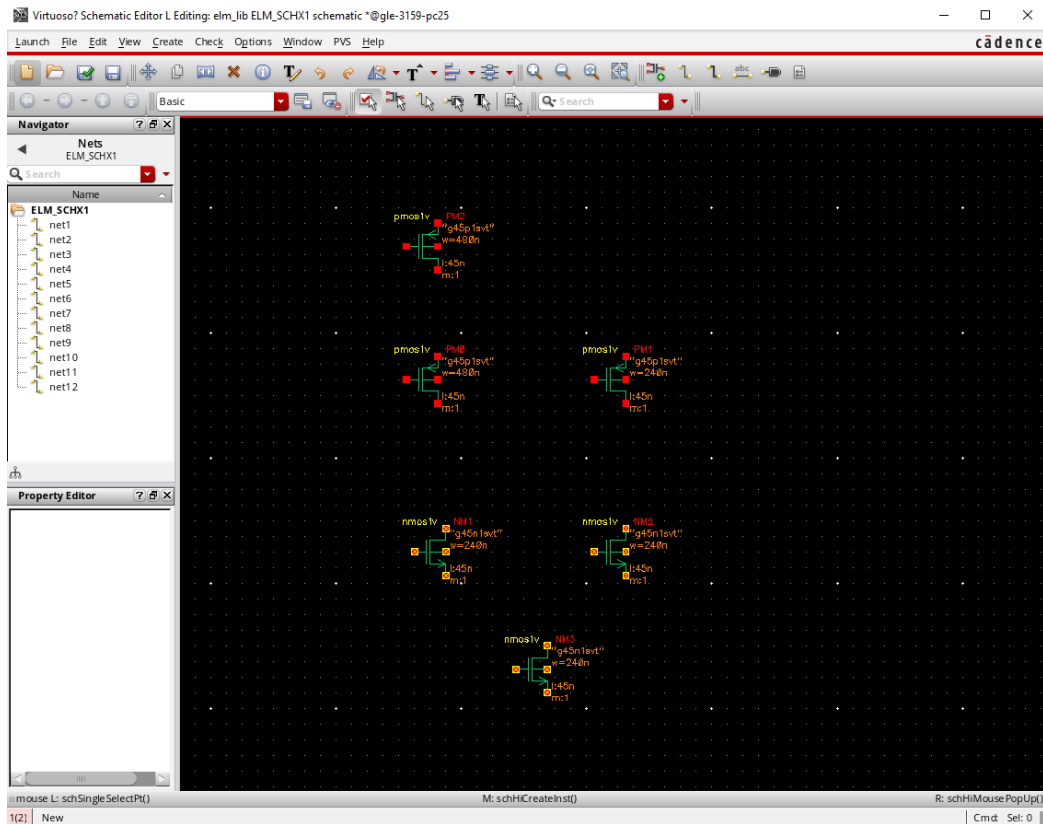


Figure 16: All Transistors in Place

OPTIONAL: If you'd like to have the transistor face a different direction, use the **Flip** tab and choose flip Horizontal (see Fig. 17). You will be on Flip Horizontal mode, so just click on the transistor of your choice.

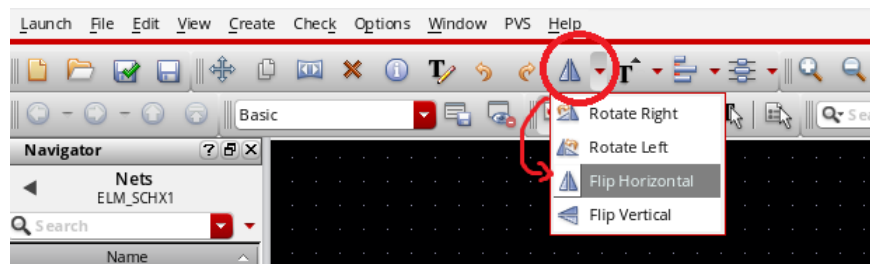


Figure 17: Flip button

Select **Create → Wire** from the menu of icons. Make connections as shown in Fig. 18. Click once for the starting point of the wire, and one more time for the end point. Notice that it is sufficient to be close to the desired and connection point and the Cadence tools will make the right connection. At the end, make sure you press ESC, otherwise the current command will remain selected. The current active command is always shown in the status bar.

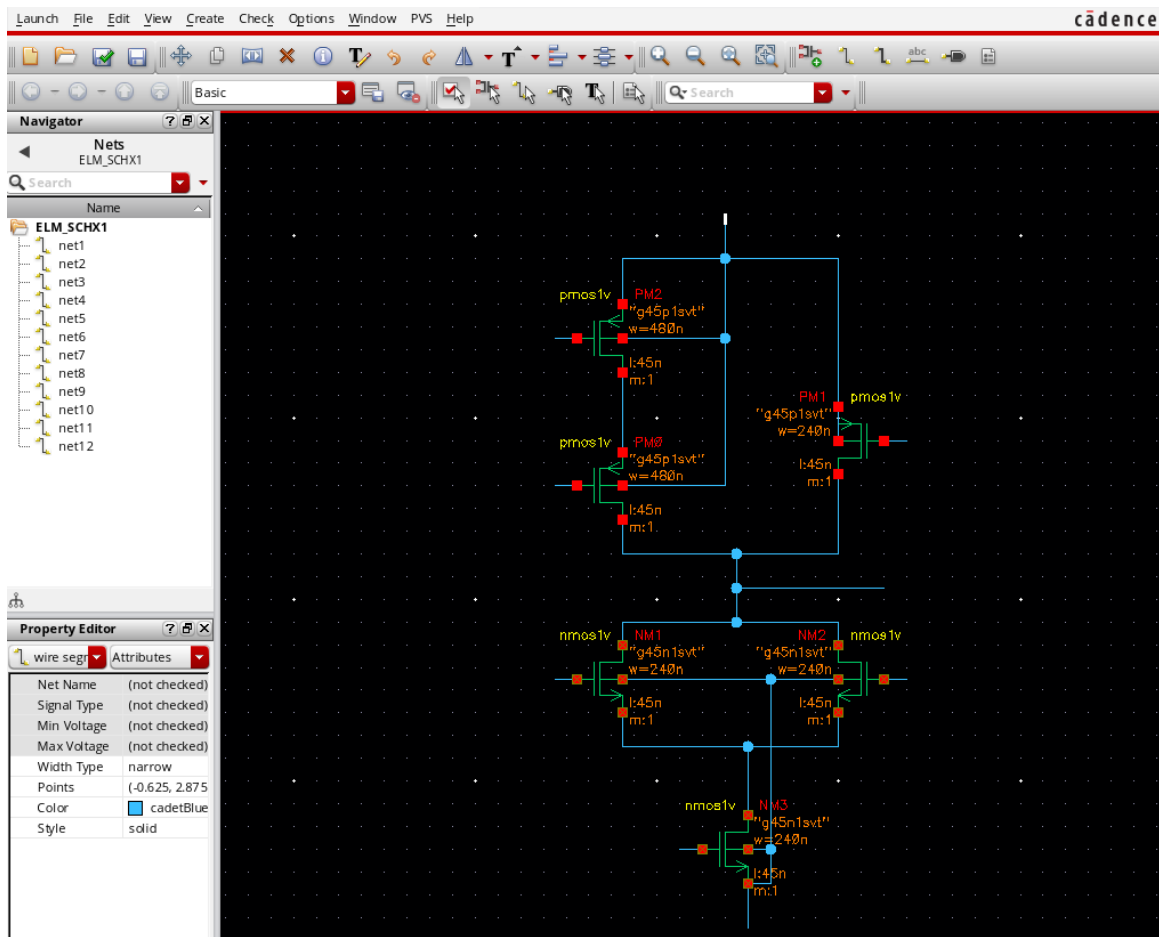


Figure 18: Initial Schematic Wiring

You may use the undo button located a button away on the left side of the **flip** button. A handy hot key, you may use your keyboard and press **U** to also undo mistakes.

Please remember to connect your body connections. This is important, otherwise your circuit will not simulate correctly!

Now **save** your work. Click on the **Check and Save** icon (the floppy disk icon with the green check mark on top). At this time, you will get some warnings. You can ignore them just for now.

REMEMBER to always save your work frequently. The lab will not remind you to save your work in the future.

Let's create some pins! Select **Create → Pin**. In the window that pops up (shown in Fig. 19) type **A** for the name of the input pin and select direction as **input** (this might be default on the menu).

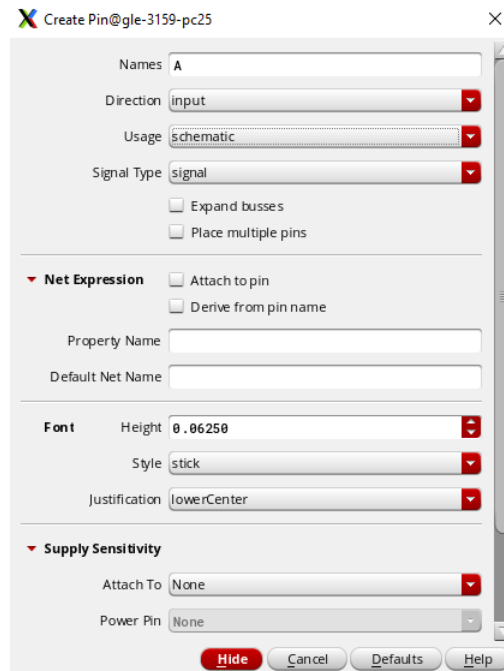


Figure 19: Add Input Pin Form

Then click **Hide**, and a yellow input pin symbol will show up attached to the cursor in the capture window.

You do: First, attach input pin A to the gates as shown in Fig 20. Create and attach the remaining pins (B,C, and Y) as well. Fig. 21 demonstrates how an output pin can be created. After doing so, your schematic should be identical to the schematic in Fig. 20.

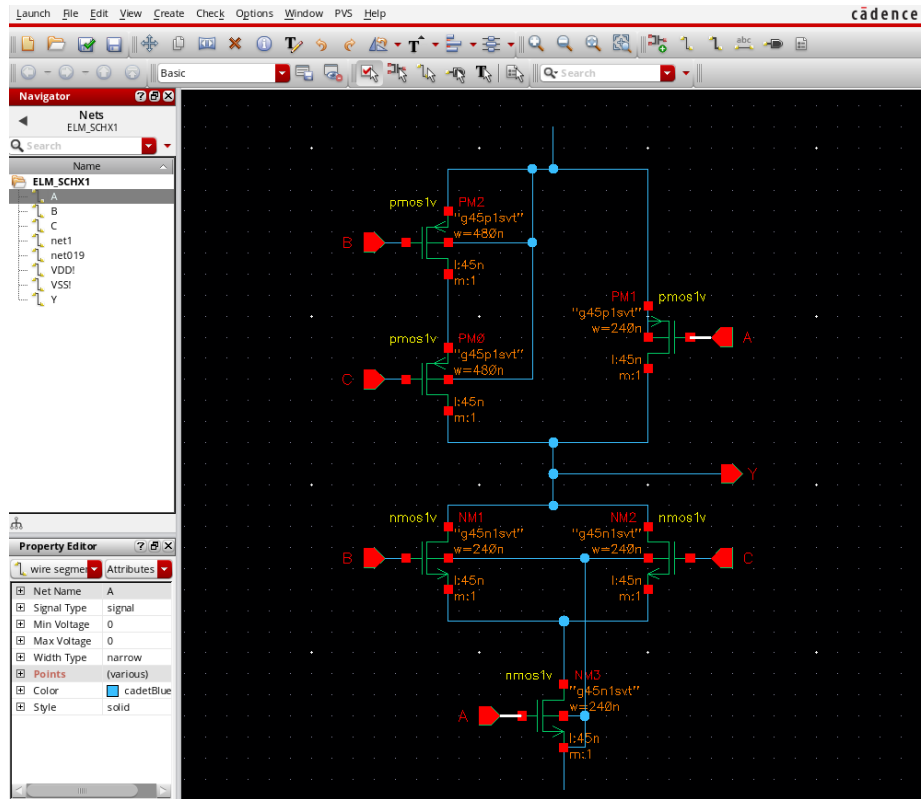


Figure 20: Schematic with I/O Pins

Figure 21: Add Output Pin Form

Figure 22: Add VDD Pin Form

One more thing left, we just need to add **VDD** and **VSS** power pins. Follow the same procedure

for creating an input pin, **BUT** make sure the signal input is **power**. Note that for power pins, you must add a Net Expression as shown in Fig. 22 for the VDD pin.

Now follow the same convention and add the VSS pin - note that since VSS is **ground** the signal type will be **ground**.

Refer to Fig. 23 for the placement of the VDD and VSS pin.

Check and save (floppy disk with a check icon, it is right below Edit) your schematic and check the CIW for log messages that your schematic is clean. You should receive a **schematic check completed with no errors** (see Fig. 23).

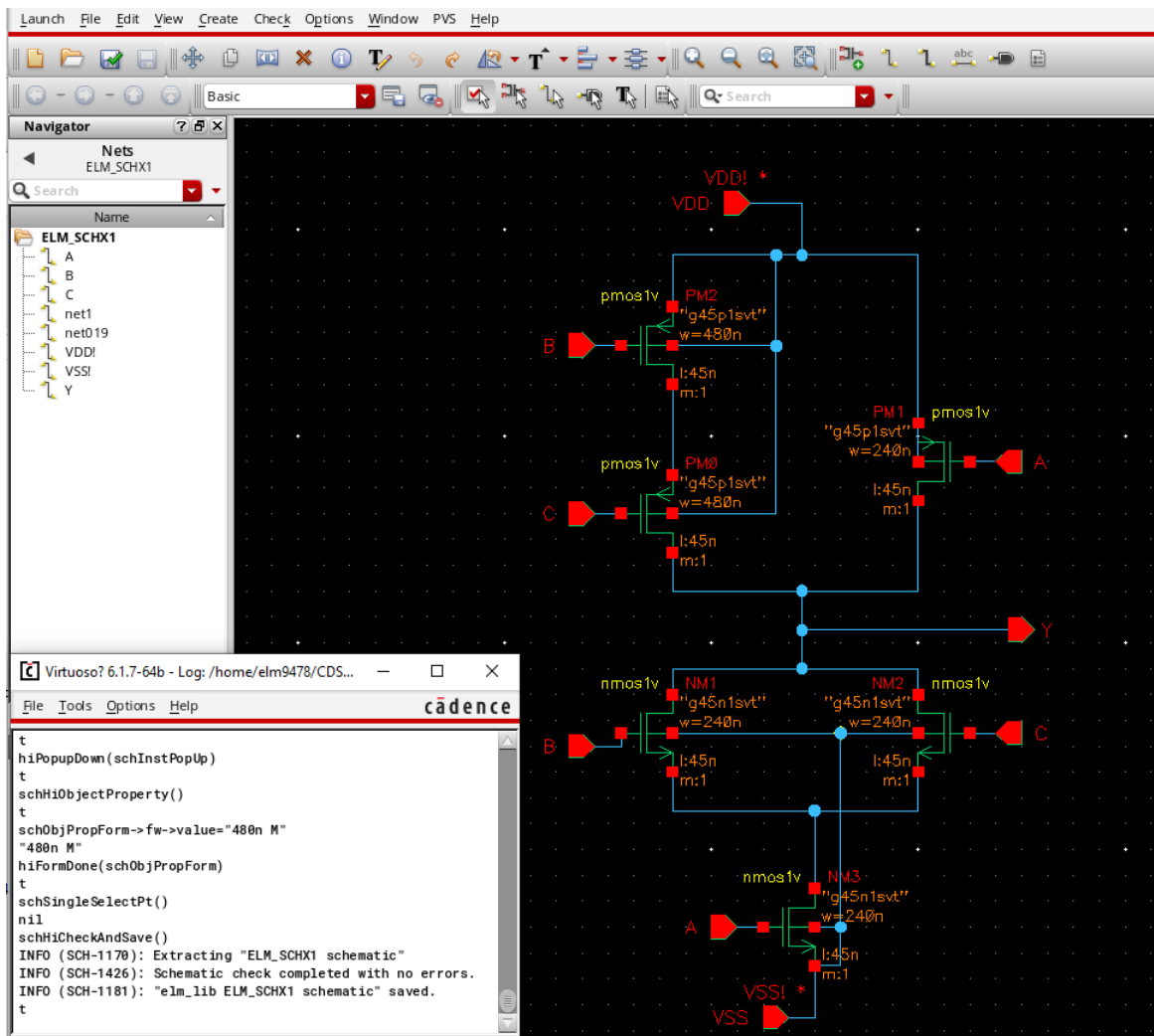


Figure 23: CLEAN final schematic with no errors message

Voila! You now know how to make a schematic with the Cadence tools! Please show your **schematic**, **CIW log**, and the **worksheet** (see next page) to the TAs for a sign-off.

6 Worksheet

Name:

Date:

1. Congratulations on finishing your very first schematic using the Cadence tools! Now evaluate the circuit. Then, write the boolean expression AND draw the logic gate circuit equivalent of the schematic you just created.

Task	Point Value	Points Earned	Date
Final Schematic & CIW Window	90		
Boolean Expression	5		
Logic Gate Circuit	5		
Total	100		